

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from sklearn.model_selection import train_test_split, GridSearchCV

from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import StandardScaler
```

```
In [2]: # 1. Data importing
df=pd.read_csv('brain_stroke.csv')

df.head()
```

```
Out[2]:
```

	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke
0	Male	67.0	0	1	Yes	Private	Urban	228.69	36.6	formerly smoked	1
1	Male	80.0	0	1	Yes	Private	Rural	105.92	32.5	never smoked	1
2	Female	49.0	0	0	Yes	Private	Urban	171.23	34.4	smokes	1
3	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.0	never smoked	1
4	Male	81.0	0	0	Yes	Private	Urban	186.21	29.0	formerly smoked	1

```
In [12]: # 2. Extracting details of data
df.shape
```

```
Out[12]: (4981, 11)
```

```
In [ ]: df.columns
```

```
In [ ]: df.info()
```

```
In [3]: df.isnull().sum().sum()
```

```
Out[3]: 0
```

```
In [ ]: #3. Data cleaning  
# As data is not having null values to handle
```

```
In [ ]: # 4. EDA  
  
#     4.1.1 Checking data inconsistency
```

```
In [4]: print(df['gender'].unique())  
  
['Male' 'Female']
```

```
In [5]: print(df['gender'].value_counts())  
  
Female    2907  
Male      2074  
Name: gender, dtype: int64
```

```
In [ ]: print(df['stroke'].value_counts())
```

```
In [ ]: print(df['stroke'].value_counts(normalize = True))
```

```
In [6]: for col in df.columns:  
        print("{} ".format(df[col].value_counts()))
```

```

Female      2907
Male        2074
Name: gender, dtype: int64
78.00      102
57.00       92
54.00       85
51.00       84
79.00       84
...
1.16        3
0.48        3
0.40        2
0.08        2
0.16        1
Name: age, Length: 104, dtype: int64
0      4502
1       479
Name: hypertension, dtype: int64
0      4706
1       275
Name: heart_disease, dtype: int64
Yes     3280
No      1701
Name: ever_married, dtype: int64
Private      2860
Self-employed    804
children       673
Govt_job       644
Name: work_type, dtype: int64
Urban      2532
Rural      2449
Name: Residence_type, dtype: int64
93.88       6
73.00       5
83.16       5
91.85       5
72.49       5
..
130.56      1
67.78       1
116.12      1
59.63       1

```

```

83.94      1
Name: avg_glucose_level, Length: 3895, dtype: int64
28.7       42
28.4       41
27.3       38
27.7       37
26.7       37
..
48.4       1
48.2       1
46.6       1
40.6       1
14.9       1
Name: bmi, Length: 342, dtype: int64
never smoked      1838
Unknown           1500
formerly smoked    867
smokes            776
Name: smoking_status, dtype: int64
0      4733
1       248
Name: stroke, dtype: int64

```

```
In [ ]: sns.barplot(x = 'gender',y = 'stroke',data=df)
```

```
In [ ]: sns.barplot(x = 'work_type',y = 'stroke',data=df)
plt.xticks(rotation=90)
plt.show()
```

```
In [ ]: sns.barplot(x = 'ever_married',y = 'stroke',data=df)
plt.xticks(rotation=90)
plt.show()
```

```
In [ ]: sns.barplot('Residence_type','stroke',data=df)
plt.xticks(rotation=90)
plt.show()
```

```
In [ ]: df.info()
```

```
In [ ]: # feature engineering
# handling categorical data
```

```
In [7]: categorial_clmn=['gender','ever_married','work_type','Residence_type','smoking_status']
# 'gender', 'age', 'hypertension', 'heart_disease', 'ever_married',
# 'work_type', 'Residence_type', 'avg_glucose_level', 'bmi',
# 'smoking_status', 'stroke'
```

```
In [8]: # df_new=pd.get_dummies(df.[categorial_clmn],drop_first=True)
df_new=pd.get_dummies(df, columns=['gender','ever_married','work_type','Residence_type','smoking_status'])
df_new.head()
```

```
Out[8]:
```

	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke	gender_Female	gender_Male	ever_married_No	ever_married_Yes	wo
0	67.0	0	1	228.69	36.6	1	0	1	0	1	
1	80.0	0	1	105.92	32.5	1	0	1	0	1	
2	49.0	0	0	171.23	34.4	1	1	0	0	1	
3	79.0	1	0	174.12	24.0	1	1	0	0	1	
4	81.0	0	0	186.21	29.0	1	0	1	0	1	

```
In [11]: len(df_new.columns)
```

```
Out[11]: 20
```

```
In [13]: # data preparation
x=df_new.drop('stroke',axis=1)
y=df_new['stroke']
x
```

Out[13]:

	age	hypertension	heart_disease	avg_glucose_level	bmi	gender_Female	gender_Male	ever_married_No	ever_married_Yes	work_ty
0	67.0	0	1	228.69	36.6	0	1	0	1	
1	80.0	0	1	105.92	32.5	0	1	0	1	
2	49.0	0	0	171.23	34.4	1	0	0	1	
3	79.0	1	0	174.12	24.0	1	0	0	1	
4	81.0	0	0	186.21	29.0	0	1	0	1	
...	...	...	...	...	...	...	...	...	...	...
4976	41.0	0	0	70.15	29.8	0	1	1	0	
4977	40.0	0	0	191.15	31.1	0	1	0	1	
4978	45.0	1	0	95.02	31.8	1	0	0	1	
4979	40.0	0	0	83.94	30.0	0	1	0	1	
4980	80.0	1	0	83.75	29.1	1	0	0	1	

4981 rows × 19 columns

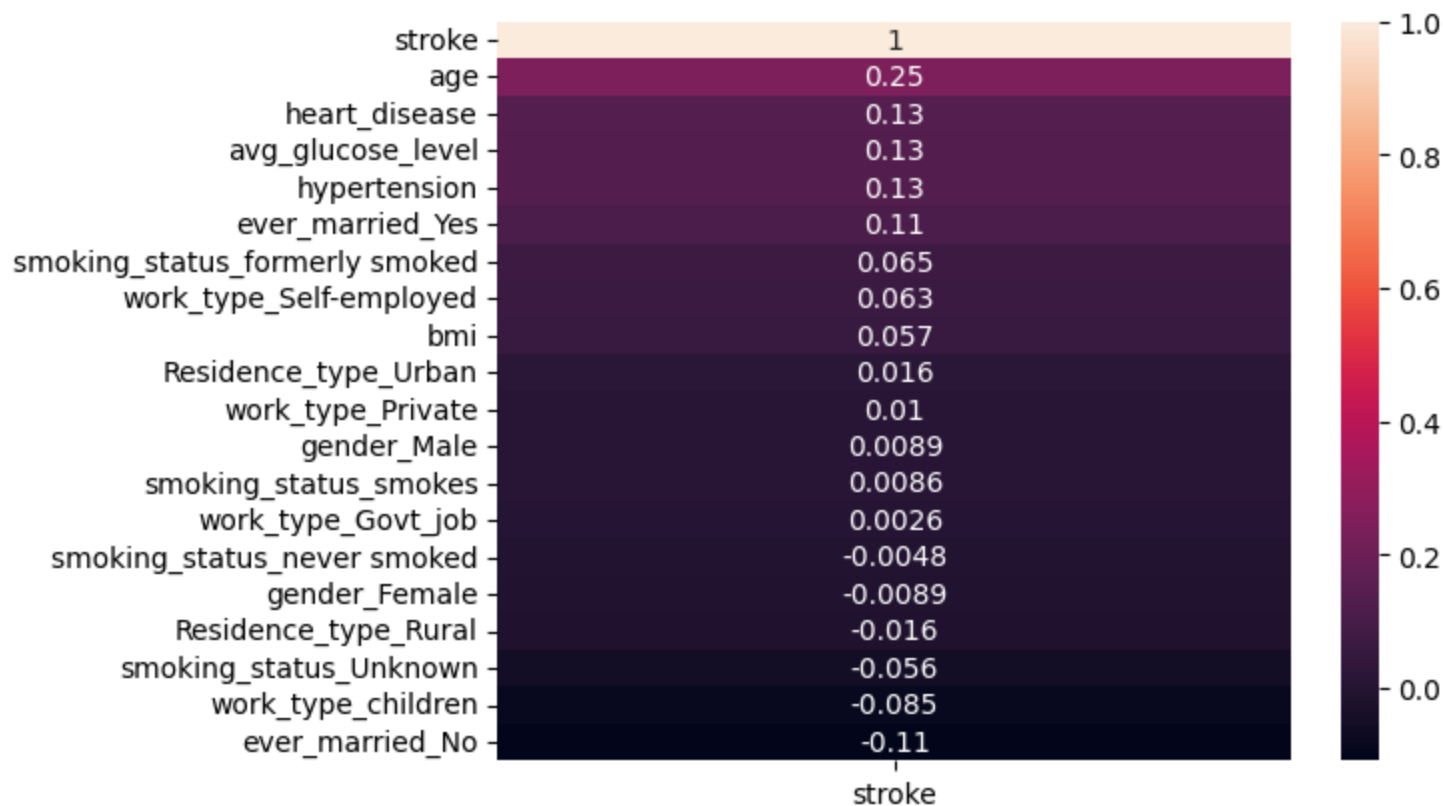
```
In [24]: # splitting data
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.25,random_state=23, stratify = y)
```

```
In [ ]: print(x_train.shape)
print(x_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
In [15]: def model_evl(model_name,actual,pred):
acc=accuracy_score(actual,pred)
print(f" Accuracy_score of {model_name} is {acc}")
```

```
In [19]: sns.heatmap(df_new.corr()[['stroke']].sort_values(by = 'stroke', ascending = False),annot = True)
```

Out[19]: <AxesSubplot: >



In [20]: `y_train.value_counts()`

Out[20]: 0 3547
1 188
Name: stroke, dtype: int64

Import DecisionTree Model

In [38]: `from sklearn.tree import DecisionTreeClassifier, plot_tree
DT=DecisionTreeClassifier(max_depth = 5,criterion = "entropy")
DT=DecisionTreeClassifier()`

In [39]: `DT.fit(x_train,y_train)`

```
Out[39]: ▾ DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy', max_depth=5)
```

```
In [40]: y_pred=DT.predict(x_test)
y_pred_train=DT.predict(x_train)
print(f" Training Accuracy =",end=" ")
model_ev1('DecisionTree',y_train,y_pred_train,)
print(f" Testing Accuracy =",end=" ")
model_ev1('DecisionTree',y_test,y_pred,)
print(f"Confusion Matirx of DT is \n {confusion_matrix(y_test,y_pred)}")
print(f"Classification Report of DT is \n {classification_report(y_test,y_pred)}")
```

Training Accuracy = Accuracy_score of DecisionTree is 0.951004016064257

Testing Accuracy = Accuracy_score of DecisionTree is 0.9486356340288925

Confusion Matirx of DT is

```
[[1182   2]
 [  62   0]]
```

Classification Report of DT is

	precision	recall	f1-score	support
0	0.95	1.00	0.97	1184
1	0.00	0.00	0.00	62
accuracy			0.95	1246
macro avg	0.48	0.50	0.49	1246
weighted avg	0.90	0.95	0.93	1246

```
In [41]: DT.score(x_train,y_train)
accuracy_score(y_pred,y_test)
```

```
Out[41]: 0.9486356340288925
```

```
In [ ]: from sklearn import tree
plt.figure(figsize=(15,10))
tree.plot_tree(DT,filled=True)
```

Hyper parameter tuning

```
In [28]: param_grid_dt={'criterion':['gini','entropy'],'max_depth':np.arange(3,20)}
```

```
In [29]: DT_grid=GridSearchCV(DT,param_grid_dt,refit=True,verbose=3)
```

```
In [30]: DT_grid.fit(x_train,y_train)
```

Fitting 5 folds for each of 34 candidates, totalling 170 fits

```
[CV 1/5] END .....criterion=gini, max_depth=3;, score=0.949 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=3;, score=0.950 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=3;, score=0.949 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=3;, score=0.950 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=3;, score=0.948 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=4;, score=0.949 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=4;, score=0.946 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=4;, score=0.948 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=4;, score=0.950 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=4;, score=0.944 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=5;, score=0.942 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=5;, score=0.946 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=5;, score=0.945 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=5;, score=0.949 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=5;, score=0.942 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=6;, score=0.941 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=6;, score=0.948 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=6;, score=0.944 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=6;, score=0.949 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=6;, score=0.944 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=7;, score=0.934 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=7;, score=0.940 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=7;, score=0.928 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=7;, score=0.940 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=7;, score=0.933 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=8;, score=0.933 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=8;, score=0.942 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=8;, score=0.936 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=8;, score=0.941 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=8;, score=0.933 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=9;, score=0.932 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=9;, score=0.934 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=9;, score=0.929 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=9;, score=0.928 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=9;, score=0.930 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=10;, score=0.920 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=10;, score=0.934 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=10;, score=0.921 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=10;, score=0.929 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=10;, score=0.926 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=11;, score=0.921 total time= 0.0s
```

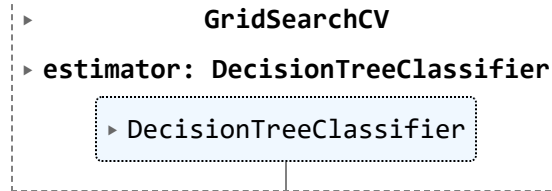
```
[CV 2/5] END .....criterion=gini, max_depth=11;, score=0.917 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=11;, score=0.921 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=11;, score=0.925 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=11;, score=0.921 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=12;, score=0.914 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=12;, score=0.898 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=12;, score=0.918 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=12;, score=0.925 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=12;, score=0.910 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=13;, score=0.908 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=13;, score=0.914 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=13;, score=0.920 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=13;, score=0.925 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=13;, score=0.914 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=14;, score=0.909 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=14;, score=0.909 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=14;, score=0.912 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=14;, score=0.920 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=14;, score=0.904 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=15;, score=0.913 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=15;, score=0.898 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=15;, score=0.913 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=15;, score=0.921 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=15;, score=0.902 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=16;, score=0.905 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=16;, score=0.896 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=16;, score=0.912 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=16;, score=0.925 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=16;, score=0.909 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=17;, score=0.906 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=17;, score=0.897 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=17;, score=0.910 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=17;, score=0.913 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=17;, score=0.908 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=18;, score=0.902 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=18;, score=0.896 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=18;, score=0.909 total time= 0.0s
[CV 4/5] END .....criterion=gini, max_depth=18;, score=0.925 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=18;, score=0.909 total time= 0.0s
[CV 1/5] END .....criterion=gini, max_depth=19;, score=0.901 total time= 0.0s
[CV 2/5] END .....criterion=gini, max_depth=19;, score=0.889 total time= 0.0s
[CV 3/5] END .....criterion=gini, max_depth=19;, score=0.910 total time= 0.0s
```

```
[CV 4/5] END .....criterion=gini, max_depth=19;, score=0.914 total time= 0.0s
[CV 5/5] END .....criterion=gini, max_depth=19;, score=0.902 total time= 0.0s
[CV 1/5] END ....criterion=entropy, max_depth=3;, score=0.949 total time= 0.0s
[CV 2/5] END ....criterion=entropy, max_depth=3;, score=0.950 total time= 0.0s
[CV 3/5] END ....criterion=entropy, max_depth=3;, score=0.950 total time= 0.0s
[CV 4/5] END ....criterion=entropy, max_depth=3;, score=0.950 total time= 0.0s
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[CV 1/5] END ....criterion=entropy, max_depth=4;, score=0.949 total time= 0.0s
[CV 2/5] END ....criterion=entropy, max_depth=4;, score=0.952 total time= 0.0s
[CV 3/5] END ....criterion=entropy, max_depth=4;, score=0.950 total time= 0.0s
[CV 4/5] END ....criterion=entropy, max_depth=4;, score=0.950 total time= 0.0s
[CV 5/5] END ....criterion=entropy, max_depth=4;, score=0.948 total time= 0.0s
[CV 1/5] END ....criterion=entropy, max_depth=5;, score=0.946 total time= 0.0s
[CV 2/5] END ....criterion=entropy, max_depth=5;, score=0.952 total time= 0.0s
[CV 3/5] END ....criterion=entropy, max_depth=5;, score=0.946 total time= 0.0s
[CV 4/5] END ....criterion=entropy, max_depth=5;, score=0.950 total time= 0.0s
[CV 5/5] END ....criterion=entropy, max_depth=5;, score=0.942 total time= 0.0s
[CV 1/5] END ....criterion=entropy, max_depth=6;, score=0.938 total time= 0.0s
[CV 2/5] END ....criterion=entropy, max_depth=6;, score=0.949 total time= 0.0s
[CV 3/5] END ....criterion=entropy, max_depth=6;, score=0.945 total time= 0.0s
[CV 4/5] END ....criterion=entropy, max_depth=6;, score=0.944 total time= 0.0s
[CV 5/5] END ....criterion=entropy, max_depth=6;, score=0.944 total time= 0.0s
[CV 1/5] END ....criterion=entropy, max_depth=7;, score=0.940 total time= 0.0s
[CV 2/5] END ....criterion=entropy, max_depth=7;, score=0.949 total time= 0.0s
[CV 3/5] END ....criterion=entropy, max_depth=7;, score=0.933 total time= 0.0s
[CV 4/5] END ....criterion=entropy, max_depth=7;, score=0.948 total time= 0.0s
[CV 5/5] END ....criterion=entropy, max_depth=7;, score=0.945 total time= 0.0s
[CV 1/5] END ....criterion=entropy, max_depth=8;, score=0.940 total time= 0.0s
[CV 2/5] END ....criterion=entropy, max_depth=8;, score=0.934 total time= 0.0s
[CV 3/5] END ....criterion=entropy, max_depth=8;, score=0.933 total time= 0.0s
[CV 4/5] END ....criterion=entropy, max_depth=8;, score=0.937 total time= 0.0s
[CV 5/5] END ....criterion=entropy, max_depth=8;, score=0.937 total time= 0.0s
[CV 1/5] END ....criterion=entropy, max_depth=9;, score=0.942 total time= 0.0s
[CV 2/5] END ....criterion=entropy, max_depth=9;, score=0.938 total time= 0.0s
[CV 3/5] END ....criterion=entropy, max_depth=9;, score=0.924 total time= 0.0s
[CV 4/5] END ....criterion=entropy, max_depth=9;, score=0.934 total time= 0.0s
[CV 5/5] END ....criterion=entropy, max_depth=9;, score=0.930 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=10;, score=0.926 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=10;, score=0.917 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=10;, score=0.918 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=10;, score=0.932 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=10;, score=0.933 total time= 0.0s
```

```
[CV 1/5] END ...criterion=entropy, max_depth=11;, score=0.932 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=11;, score=0.913 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=11;, score=0.918 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=11;, score=0.929 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=11;, score=0.921 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=12;, score=0.921 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=12;, score=0.906 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=12;, score=0.922 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=12;, score=0.925 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=12;, score=0.917 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=13;, score=0.917 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=13;, score=0.904 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=13;, score=0.913 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=13;, score=0.909 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=13;, score=0.918 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=14;, score=0.916 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=14;, score=0.900 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=14;, score=0.910 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=14;, score=0.912 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=14;, score=0.917 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=15;, score=0.914 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=15;, score=0.900 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=15;, score=0.909 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=15;, score=0.913 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=15;, score=0.910 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=16;, score=0.909 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=16;, score=0.901 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=16;, score=0.901 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=16;, score=0.905 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=16;, score=0.908 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=17;, score=0.914 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=17;, score=0.904 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=17;, score=0.900 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=17;, score=0.902 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=17;, score=0.906 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=18;, score=0.914 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=18;, score=0.900 total time= 0.0s
[CV 3/5] END ...criterion=entropy, max_depth=18;, score=0.905 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=18;, score=0.905 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=18;, score=0.914 total time= 0.0s
[CV 1/5] END ...criterion=entropy, max_depth=19;, score=0.920 total time= 0.0s
[CV 2/5] END ...criterion=entropy, max_depth=19;, score=0.898 total time= 0.0s
```

```
[CV 3/5] END ...criterion=entropy, max_depth=19;, score=0.905 total time= 0.0s
[CV 4/5] END ...criterion=entropy, max_depth=19;, score=0.904 total time= 0.0s
[CV 5/5] END ...criterion=entropy, max_depth=19;, score=0.908 total time= 0.0s
```

Out[30]:



In [77]: DT_grid.best_params_

Out[77]: {'criterion': 'entropy', 'max_depth': 3}

```
In [78]: y_pred=DT_grid.predict(x_test)
y_pred_train=DT_grid.predict(x_train)
print(f" Training Accuracy =",end=" ")
model_ev1('DecisionTree after parameter tuning',y_train,y_pred_train,)
print(f" Testing Accuracy =",end=" ")
model_ev1('DecisionTree after parameter tuning',y_test,y_pred,)
print(f"Confusion Matirx of DT after parameter tuning is \n {confusion_matrix(y_test,y_pred)}")
print(f"Classification Report of DT after parameter tuning is \n {classification_report(y_test,y_pred)}")
```

Training Accuracy = Accuracy_score of DecisionTree after parameter tuning is 0.9502008032128514

Testing Accuracy = Accuracy_score of DecisionTree after parameter tuning is 0.9502407704654896

Confusion Matirx of DT after parameter tuning is

```
[[1184    0]
 [  62    0]]
```

Classification Report of DT after parameter tuning is

	precision	recall	f1-score	support
0	0.95	1.00	0.97	1184
1	0.00	0.00	0.00	62
accuracy			0.95	1246
macro avg	0.48	0.50	0.49	1246
weighted avg	0.90	0.95	0.93	1246

```
C:\Users\MADHAV\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics\_classification.py:1344: UndefinedMetricWarning: Precision and F-score are ill-defined and being set to 0.0 in labels with no predicted sample s. Use `zero_division` parameter to control this behavior.
_warn_prf(average, modifier, msg_start, len(result))
C:\Users\MADHAV\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics\_classification.py:1344: UndefinedMetricWarning: Precision and F-score are ill-defined and being set to 0.0 in labels with no predicted sample s. Use `zero_division` parameter to control this behavior.
_warn_prf(average, modifier, msg_start, len(result))
C:\Users\MADHAV\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\metrics\_classification.py:1344: UndefinedMetricWarning: Precision and F-score are ill-defined and being set to 0.0 in labels with no predicted sample s. Use `zero_division` parameter to control this behavior.
_warn_prf(average, modifier, msg_start, len(result))
```

```
In [79]: # Model is regularized now after parameter tuning using gridsearch classifier
```

```
In [80]: DT_improved=DecisionTreeClassifier(criterion='entropy', max_depth= 3)
# DT_improved=DecisionTreeClassifier(criterion='entropy')
```

```
In [81]: DT_improved.fit(x_train,y_train)
```

```
Out[81]: ▼ DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy', max_depth=3)
```

```
In [82]: DT_improved.score(x_train,y_train)
```

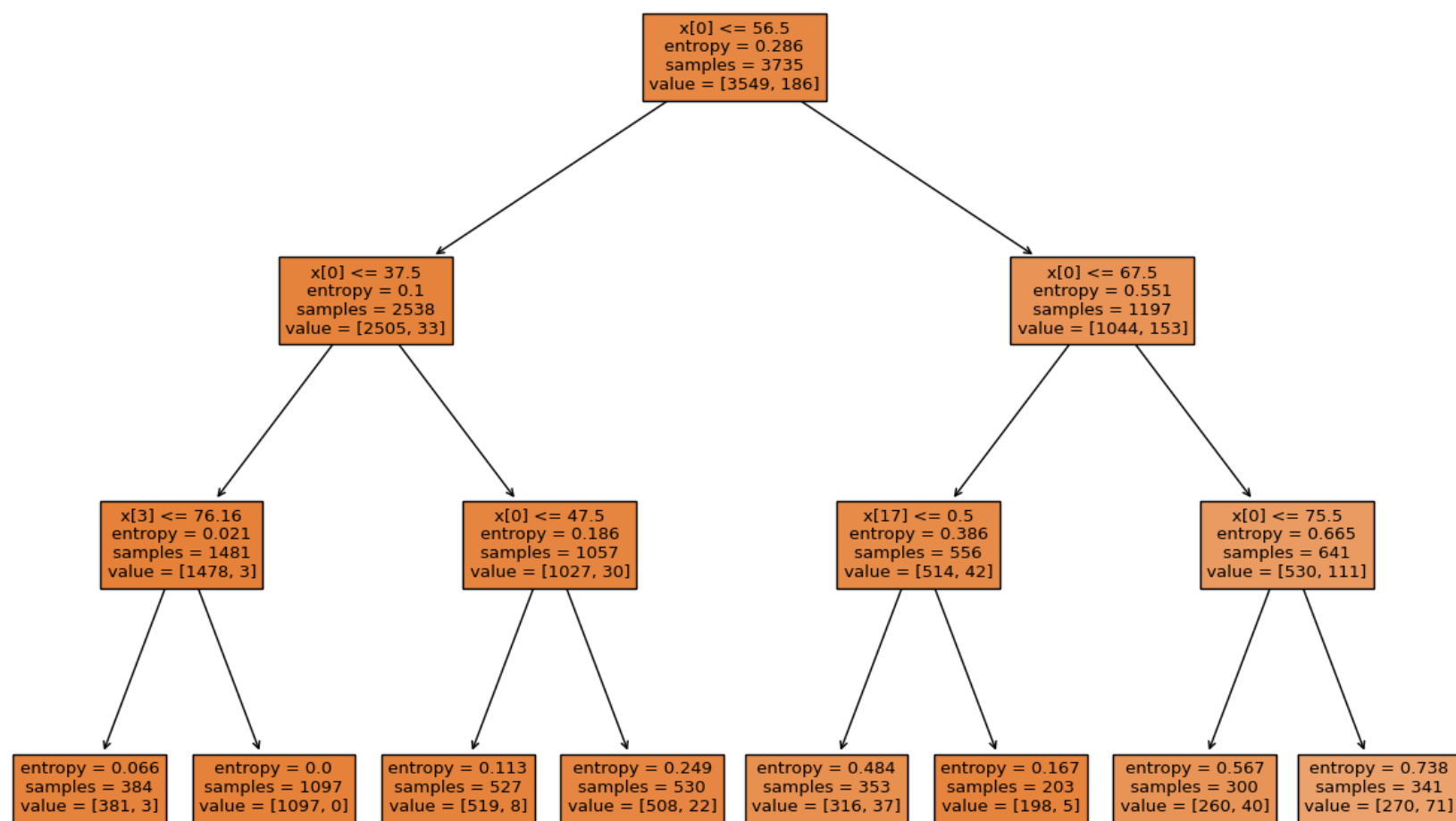
```
Out[82]: 0.9502008032128514
```

```
In [83]: y_pred_impr=DT_improved.predict(x_test)
```

```
In [84]: accuracy_score(y_pred_impr,y_test)
```

```
Out[84]: 0.9502407704654896
```

```
In [85]: plt.figure(figsize=(15,10))
plot_tree(DT_improved,filled=True)
plt.savefig('dt1')
plt.show()
```



In [87]: `from matplotlib.backends.backend_pdf import PdfPages`

```

with PdfPages('dts') as pdf:
    fig = plt.figure()
    pdf.savefig(fig)
    pdf.savefig()
  
```

<Figure size 640x480 with 0 Axes>

In []:

<https://www.geeksforgeeks.org/gini-impurity-and-entropy-in-decision-tree-ml/>

In []: